The Greeks

1) Delta (δ)

It represents how much an option value will change with \$1 increase in the stock it represents.

Call : $0 \longleftrightarrow +1$
Put : $-1 \longleftrightarrow 0$

⇒ Higher for "in the money" or profitable options

Lower for "out of the money" or unprofitable options.

⇒ Mathematically, $\Delta = \dfrac{\partial C}{\partial S}$

It is only for local changes over short period of time. (Local Approx.)

Challaenge → δ changes with Stock Price

2) Gamma (γ)

Measures the sensitivity of δ with stock price.

Mathematically,

$$\boxed{\gamma = \frac{\partial^2 C}{\partial S^2}}$$

=) γ is higher for at the money.
=) γ is lower for deep in / out of money

In the money (ITM) =) Profitable ($S > K$)

At the money (ATM) =) Almost same ($S \approx K$)

Out of the money (OTM) =) Lose money ($S < K$)

Deep in money =) Very Profitable
Deep out of money =) Far from Profitable

3) Theta (θ)

=) How much time impacts option value.
=) Every passing day reduces the value.

Buyer =) θ (-ve) Less time to make it profitable
Seller =) θ (+ve) Benefits if option is unused

Mathematically,

$$\theta = \frac{\partial C}{\partial t}$$

=) Theta is higher for at the money.
Lower for Deep In, Out of money.

4) Vega (v)

Measures volatility or unstability of stock Price.

=) More randomness is preffered over stable as it makes it more probable to hit strike Price.

Mathematically =) Implied volatility is used;

$$v = \frac{\partial C}{\partial \sigma}$$

(expected volatility)

v =) Higher for ATM, longer terms of maturity

5) Rho (ρ)

Measures change in option value with rate of interest.

+ve for Call options
-ve for Put options

Mathematically,

$$\rho = \frac{\partial C}{\partial r}$$

=) Higher for ATM, For Longer Terms

$\delta_{call} = N(d_1)$ $\delta_{put} = N(d_1) - 1$

$$\text{Gamma} = \frac{N'(d_1)}{S \cdot \sigma \cdot \sqrt{T}}$$

$$\text{Theta}_{call} = -\frac{S \cdot N'(d_1) \cdot \sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$$

$$\text{Theta}_{put} = -\frac{S \cdot N'(d_1) \cdot \sigma}{2\sqrt{T}} + rKe^{-rt}N(-d_2)$$

$$\text{Vega} = S \cdot N'(d_1) \cdot \sqrt{T}$$

$\text{Rho}_{put} = -KTe^{-rT}N(-d_2)$ $\text{Rho}_{call} = KTe^{-rT}N(d_2)$